

波动光学

一、光的电磁理论

真空光速: $C = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$

介质中的光速: $v = \frac{c}{\sqrt{\epsilon_r \mu_r}} = \frac{c}{n}$

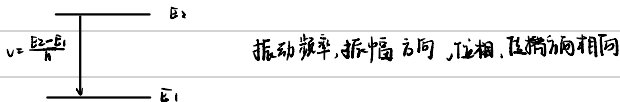
二、光源

1. 发光本质: 原子能级及发光跃迁

2. (1) 普通光源 非相干光源

间歇性 随机性(独立)

(2) 激光光源: 受激辐射



3. (1) 光源振动方程: $E = E_0 \cos(\omega t + \varphi_0)$

(2) 光场中任一点的光矢量(波函数): $E = E_0 \cos(\omega t - \frac{2\pi r}{\lambda} + \varphi_0)$

(3) 点光源产生球面波 (各向同性均匀介质中):

4. $u = \frac{1}{\sqrt{\epsilon_0 \epsilon_r \mu_0 \mu_r}} = \frac{1}{\sqrt{\epsilon \mu}}$

$$\vec{S} = \vec{E} \times \vec{H}$$

$$\sqrt{\epsilon} E = \sqrt{\mu} H$$

$$S = u E E^2 = u E E_0^2 \cos^2(\omega t + \varphi - \frac{2\pi r}{\lambda})$$

光强——平均能流密度 $I = \frac{1}{T} \int_0^T S dt = \frac{1}{2} u E E_0^2 = \frac{1}{2} \sqrt{\epsilon \mu} E_0^2$

三、光波的叠加

1. 线性与叠加原理

光波在空间传播时, 相应的电磁扰动会使介质极化.

在线性介质中传播, 遵从

2. 相干叠加

$$E_1 = E_{10} \cos(\omega t + \varphi_1)$$

$$E_2 = E_{20} \cos(\omega t + \varphi_2)$$

$$\text{合成数量: } \vec{E} = \vec{E}_0 \cos(\omega t + \varphi)$$

$$E_0 = E_{10}^2 + E_{20}^2 + 2E_{10}E_{20}\cos\Delta\varphi \quad \Delta = \varphi_2 - \varphi_1$$

由所观察到的光强 I

$$I = \overline{E^2} = \frac{1}{T} \int_0^T (E_{10}^2 + E_{20}^2 + 2E_{10}E_{20}\cos\Delta\varphi) dt$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\Delta\varphi$$

1) 非相干光源

$$\overline{\cos\Delta\varphi} = \frac{1}{T} \int_0^{T+\tau} \cos\Delta\varphi dt = 0 \quad I = I_1 + I_2$$

2) 完全相干光源 ($\Delta\varphi = \text{const}$)

$$\overline{\cos\Delta\varphi} = \frac{1}{T} \int_0^{T+\tau} \cos\Delta\varphi dt = \cos\Delta\varphi$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\Delta\varphi$$

$$\Delta\varphi = \pm 2k\pi, k \in \mathbb{N} \quad \text{相长干涉 明纹} \quad I_{\max}$$

$$\Delta\varphi = \pm (2k+1)\pi, k \in \mathbb{N} \quad \text{相消干涉 暗纹} \quad I_{\min}$$

$$\text{若 } I_1 = I_2 \text{ 时 } I = 2I_1(1 + \cos\Delta\varphi) = \begin{cases} \Delta\varphi = \pm 2k\pi & I_{\max} = 4I \\ \Delta\varphi = \pm (2k+1)\pi & I_{\min} = 0 \end{cases}$$

13) 条纹衬比度 $V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, 0 \leq V \leq 1$

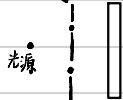
$$I_1 \neq I_2 \quad V < 1 \quad \text{衬比度差}$$

$$I_1 = I_2 \quad V = 1 \quad \text{衬比度好}$$

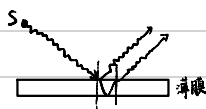
$$V = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$$

14) 普通光源获得相干光的途径

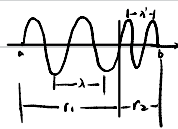
① 分波阵面法 (杨氏干涉)



② 分振幅法 (薄膜干涉)



四. 光程



$$v = \frac{c}{n}$$

$$\lambda' = \frac{v}{\nu} = \frac{c}{n\nu} = \frac{\lambda}{n}$$

$$\varphi_a + \varphi_b = \frac{2\pi r_1}{\lambda} + \frac{2\pi r_2}{\lambda} = \frac{2\pi}{\lambda} n_1 r_1 + \frac{2\pi}{\lambda} n_2 r_2 = \frac{2\pi}{\lambda} \delta \quad \delta = r_1 + n_2 r_2 = n_1 r_1 + n_2 r_2 > \text{光程}$$

1. 光在介质中任何路程算出真空中的任何路程 光程 $L = \sum n_i d_i$

2. 光程差与位相差之间的关系

$$\delta = L_2 - L_1 \quad \Delta\varphi = \frac{2\pi}{\lambda} \delta$$

$$\delta = \pm k\lambda \quad \text{明纹}$$

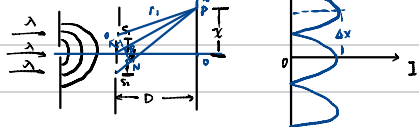
$$\delta = \pm (k + \frac{1}{2})\lambda \quad \text{暗纹}$$

3. 透镜会产生附加光程差

① 半波损失: 光疏 $\rightarrow$ 光密 $n_1 < n_2$

五. 杨氏双缝干涉

1. 基本装置



$$d > \lambda \quad D \gg d$$

2. 光程差 $\delta = r_2 - r_1 \approx \sin\theta \approx d \sin\theta$

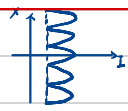
3. 屏上强弱中心位置: $\delta = d \tan\theta \approx d \cdot \frac{y}{D} = \begin{cases} \pm k\lambda & \text{明纹} \\ \pm (k + \frac{1}{2})\lambda & \text{暗纹} \end{cases}$

$$\text{明纹 } x = k\lambda \frac{D}{d} \quad \text{暗纹 } x = \pm (2k+1) \frac{D}{2d} \lambda \quad (k \in \mathbb{N}) \quad \Delta x = \frac{D}{d} \lambda$$

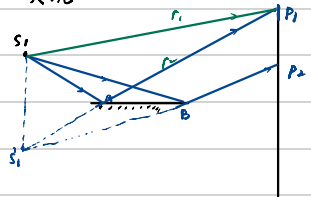
4. 光强公式

$$I = I_2 \quad I = 4I_0 \cos^2 \frac{\Delta\varphi}{2}$$

把一条缝堵上, 条纹如何变化
若 d 不变, 则条纹位置不变。
 $E_1 + E_2 = 0$ 则暗纹强度不为 0, 明纹强度也变, 条纹反差不彻底,



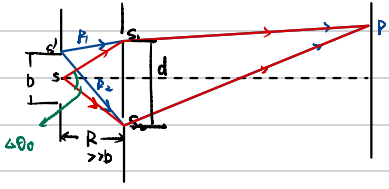
六. 洛埃镜



$$\text{真空中 } r_2 - r_1 + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明纹} \\ (2k+1)\lambda & \text{暗纹} \end{cases}$$

半波损失

(2) 空间相干性



附加光程差 δb

$$\delta b = R_2 - R_1 = \frac{R_2^2 - R_1^2}{R_2 + R_1} \approx \frac{R_2^2 - R_1^2}{2R} = \frac{1 \cdot (\frac{d}{2} + \frac{b}{2})^2 - 1 \cdot (\frac{d}{2} - \frac{b}{2})^2}{2R} = \frac{bd}{2R}$$

若 $\delta b = \frac{\lambda}{2}$, 则 S 与 S' 各自发出的光经干涉后明暗正好相反, 条纹消失, 衬比度为 0.

d - 定时, 光源的极限宽度 $\delta b < \frac{\lambda}{2}$ 即 $b < \frac{\lambda}{d}$

b - 定时, 相干范围的横向线度: $d_{max} = \frac{R}{b} \lambda$

$$\text{孔径角: } \Delta\theta_0 \approx \frac{d_{max}}{R} = \frac{\lambda}{b}$$

光源的线宽度越小, 孔径角越大, 空间相干越好

七. (1) 时间相干性

原子发光是间歇性的, 每个波列持续的时间是 $\Delta t = 10^{-8}s$

$$\delta = \begin{cases} < L & \text{发生干涉} \\ = L & \text{非相干叠加} \\ > L & \text{不发生干涉} \end{cases}$$

单色性越好, 即频率越窄
波列就越长, 相干性越好

八、分振幅干涉

11) 等倾干涉——厚度均匀的薄膜所得到的干涉

设薄膜厚度为 d , 折射率为 n

并且 $n_1 < n < n_2$

$$\delta = n(AC + BC) - n_1 AD$$

$$AC = BC = \frac{d}{\cos r}$$

$$AD = AB \sin i = 2d \tan r \sin i$$

$$\delta = \frac{2nd}{\cos r} - 2nd \sin i \tan r$$

$$\left(\begin{aligned} n_1 \sin i &= n \sin r \\ \delta &= 2d \sqrt{n^2 - n_1^2 \sin^2 i} = \begin{cases} k\lambda, & k \in \mathbb{N}^+ & \text{明纹} \\ 2k + \frac{1}{2}, & k \in \mathbb{N} & \text{暗纹} \end{cases} \end{aligned} \right)$$

$n > n_1 > n_2$ } 不考虑半波损失
 $n_1 < n < n_2$

$n_1 > n < n_2$ 考虑半波损失

特征: ① 相同 i 的光线对应同一条干涉圆环纹

② 不同 i 构成的等倾干涉是系列同心圆纹

③ 愈往中心, 条纹级次愈高

$$2d \sqrt{n^2 - n_1^2 \sin^2 i} = k\lambda$$

d 一定时, $k \uparrow \rightarrow i \rightarrow r \downarrow$

中心 O 点处的干涉级次最高

若改变 d — $d \uparrow$ 中心向外冒条纹

$d \downarrow$ 中心向内吞条纹

④ 条纹间隔分布: 内疏外密

$$\left. \begin{aligned} 2nd \cos r_k &= k\lambda \\ 2nd \cos r_{k+1} &= (k+1)\lambda \end{aligned} \right\} \Rightarrow \Delta r_k = \frac{\lambda}{2nd \sin r_k} \quad r_k \uparrow \Rightarrow \Delta r_k \downarrow$$

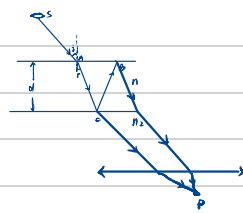
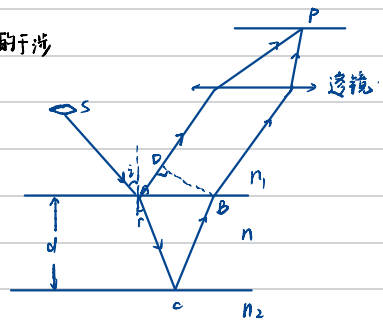
$$\cos(r_{k+1}) = \cos(r_k + \Delta r_k) \approx \cos r_k - \Delta r_k \sin r_k \quad \Delta r_k \sim 0$$

⑤ 白光入射:

$\lambda \uparrow \rightarrow i \downarrow \rightarrow r \downarrow$ 彩色干涉条纹

说明: 11) 透射光也有干涉现象

明暗条件: $2d \sqrt{n^2 - n_1^2 \sin^2 i} = \pm \frac{\lambda}{2}$

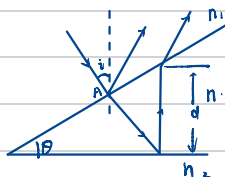


12) 等厚干涉 —— 厚度不均匀的薄膜干涉

① 劈尖干涉 (空气隙劈尖)

B处 (厚度为 d) 的明暗条件:

$$2d\sqrt{n^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2} = \begin{cases} k\lambda, & k \in \mathbb{N}^+ \text{ 明纹} \\ (2k+1)\frac{\lambda}{2}, & k \in \mathbb{N} \text{ 暗纹} \end{cases}$$



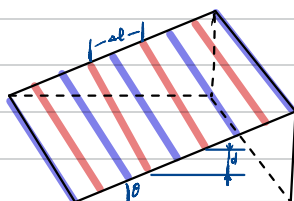
若 $i=0, n=1$

$$\delta = \Delta r = 2d + \frac{\lambda}{2} = \begin{cases} k\lambda, & k \in \mathbb{N}^+ \\ (2k+1)\frac{\lambda}{2}, & k \in \mathbb{N} \end{cases}$$

说明: ① 每一 k 值对应劈尖一确定厚度 d

即同级条纹对应同一厚度

$$\textcircled{2} \quad d=0 \quad \begin{cases} n_1 = n_2 \neq n & \text{暗纹} \\ n_1 < n < n_2 & \text{亮纹} \end{cases}$$



③ 相邻两明(暗)纹间对应的厚度差

$$\begin{aligned} 2d + \frac{\lambda}{2} &= k\lambda \\ 2(d+\Delta d) + \frac{\lambda}{2} &= (k+1)\lambda \end{aligned} \Rightarrow \Delta d = \frac{\lambda}{2} = \frac{\lambda}{2n}$$

④ 明(暗)纹间距 l :

$$l \sin \theta = \Delta d = \frac{\lambda}{2} = \frac{\lambda}{2n}$$

$$l = \frac{\lambda}{2n \sin \theta}$$

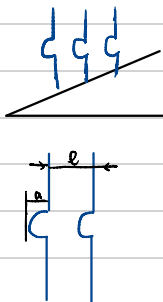
⑤ 复色光入射得彩色条纹

肥皂泡

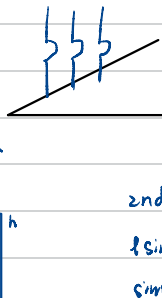
应用: ① 测量厚度 d

② 可检测工件的平整度

有“凹槽”



有“凸起”

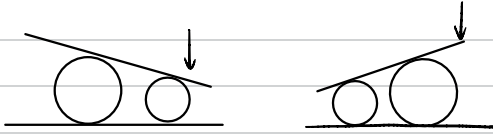


$$2nd + \frac{\lambda}{2} = k\lambda$$

$$l \sin \theta = \Delta d = \frac{\lambda}{2n}$$

$$\sin \theta = \frac{h}{a} = \frac{\lambda}{2l} \Rightarrow h = \frac{a\lambda}{2l}$$

③ 用两块平面玻璃板能否判别两个直径相差很大的圆珠



$$l = \frac{\lambda}{2n \sin \theta}$$

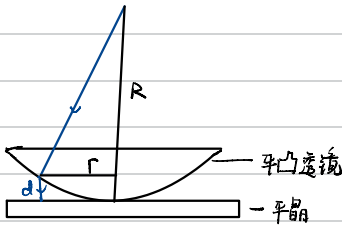
1° $\theta \uparrow \rightarrow l \downarrow \rightarrow$ 条纹变密

$\theta \downarrow \rightarrow l \uparrow \rightarrow$ 条纹变疏

2° 用白光入射 $2ndk + \frac{\lambda}{2} = k\lambda$

对同一级条纹, 形成彩带 入大! 红色 dk 大 故 总是靠近有球大的一端

② 牛顿环



明暗条件: $2d + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明} \\ (2k+1)\frac{\lambda}{2} & \text{暗} \end{cases}$

干涉环半径 $r = \begin{cases} \sqrt{\frac{12R - 11R\lambda}{2}} & k=0, 1, \dots, \text{明纹} \\ \sqrt{kR\lambda} & k=0, 1, \dots, \text{暗纹} \end{cases}$

$$r^2 = R^2 - (R-d)^2 \approx 2Rd$$

讨论: 1° $2d + \frac{\lambda}{2} = k\lambda \Rightarrow d \uparrow, k \uparrow$

2° $r_k = \sqrt{kR\lambda}$, $k \in \mathbb{N}$ 牛顿环中心是暗点

3° 相邻两暗环间隔

$$\Delta r_k = r_{k+1} - r_k \approx \sqrt{\frac{R\lambda}{k}} \quad (k \gg 1)$$

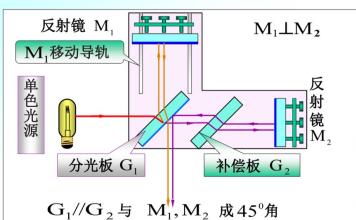
环中心疏, 内边密

4° $R = \frac{r_{k+m}^2 - r_k^2}{m\lambda}$

5° $R \rightarrow$ 增大

6° 透射光互补

③ 迈克耳逊干涉仪



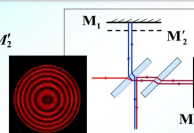
迈克耳逊干涉仪

当 $M_1 \perp M_2 \rightarrow M_1 \parallel M_2'$

M_1 与 M_2' 形成厚度

均匀的薄膜,

——等倾条纹

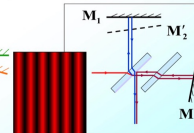


当 $M_1 \perp M_2 \rightarrow M_1 \perp M_2'$

M_1 与 M_2' 形成一空气

隙劈尖,

——等厚条纹



平移 M_2 由中央明纹满足的光程差: $\Delta r = 2d = k\lambda$

M_2 平移 $\frac{\lambda}{2} \rightarrow$ 中心有一个亮斑冒出或缩进

中心冒出或缩进的亮斑数目 $N \Rightarrow M_2$ 平移的距离 $\Delta d = N \frac{\lambda}{2}$

九 光的衍射 单缝衍射

1. 惠更斯—菲涅耳原理

波面上各面元所发出的球面子波在观察点 P 的相干叠加 决定 P 点的合振动及光强.

$$dE_{ip} \propto \frac{A(\theta)f(\theta)}{r} ds \cdot \cos(\omega t - \frac{2\pi r}{\lambda})$$

$$E_{ip} = C \int_S \frac{A(\theta)f(\theta)}{r} \cdot \cos(\omega t - \frac{2\pi r}{\lambda}) ds$$

$$= E_{op} \cdot \cos(\omega t + \varphi_p) \quad P \text{ 处光的强度 } I_p \propto |E_{op}|^2$$

衍射现象, 实为无数多个无限小的子波的干涉效应

2. 衍射光强的计算

(1) 惠更斯—菲涅耳原理

S : 单色光源

θ : 衍射角

$AB = a$ (缝宽) $\ll f$

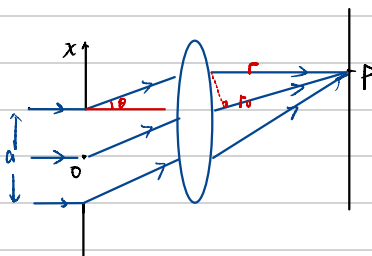
$$C \frac{A(\theta)f(\theta)}{r} \approx C'$$

$$E_{ip} = C \int_S \frac{A(\theta)f(\theta)}{r} \cos(\omega t - \frac{2\pi r}{\lambda}) ds$$

$$= C' \int_{-\frac{a}{2}}^{\frac{a}{2}} \cos(\omega t - \frac{2\pi r}{\lambda}) dx \quad ds = b dx \quad r = r_0 - x \sin \theta$$

$$= E_0 \frac{\sin \alpha}{\alpha} \cos(2\pi \frac{f}{\lambda} - \frac{\alpha}{\lambda}) \quad \text{其中: } E_0 = C'a, \quad \alpha = \frac{\pi a \sin \theta}{\lambda}$$

$$P \text{ 点合振幅为: } E_p = E_0 \frac{\sin \alpha}{\alpha} \quad P \text{ 点的光强为: } I_p = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$$



(2) 振幅矢量叠加法

将缝分成 N 个等宽窄条, 宽度 $\Delta x = \frac{a}{N}$

每窄条视为子波源, 在 P 点振动, 振幅为 ΔE_0 .

相邻两窄条波源的相位差为 $\Delta \varphi = \frac{2\pi \Delta x \sin \theta}{\lambda}$

$$P_0 \rightarrow \sin \theta = 0, \Delta \varphi = 0$$

$$I_\theta \propto E_0^2$$

$$P \rightarrow \Delta \varphi = \frac{2\pi a \sin \theta}{\lambda}$$

$$I_\theta = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

$N \rightarrow \infty$ 振幅合成圆弧

$$\text{其中 } \alpha = \frac{\Delta \varphi}{2} = \frac{\pi a \sin \theta}{\lambda}$$

$$\frac{E_p}{E_0} = \Delta \varphi = 2\alpha \Rightarrow \frac{E_p}{E_0} = \frac{2\alpha}{\alpha} = 2$$

$$E_\theta = 2R \sin \alpha = E_0 \frac{\sin \alpha}{\alpha}$$

